**FieldComms** is a self-contained emergency communications server built on a Raspberry Pi 5 for McHenry County RACES, ARES, and Starcom operations. Any smartphone, tablet, or laptop connects to the **EMCOMM-NET** Wi-Fi access point and reaches the full dashboard at **http://192.168.50.1** — no internet, no app installation, no per-device configuration required. All 30 tools, all ICS forms, all reference materials, and all offline maps remain fully functional when the site has no internet connectivity.

### AMATEUR RADIO

- Net Control Logger — FCC auto-fill
- Callsign Lookup — 800K licensees offline
- Observer Mode — read-only agency view
- APRS Tactical Map — Graywolf + YAAC
- HF Propagation — band conditions live
- Repeater Database — ARES/band filters
- Dead Man's Switch — net inactivity alert
- NTS Radiogram — ARRL formatted traffic
- JS8Call — HF keyboard digital messaging
- Pat Winlink — backup email over radio

### STARCOM / PUBLIC SAFETY

- Starcom Net Logger — Radio ID / unit-based
- Weather Net — storm spotter check-in
- SAR Net — search and rescue radio ops
- Resource Tracking Map — unit placement
- Resource Board — 5-state status cycle
- Member Roster — certifications, activations
- Facilities Directory — EOC, shelters, hospitals
- Multiple simultaneous nets on one screen
- Observer link for served agencies
- ICS platform integration from Starcom mode

### ICS INCIDENT COMMAND

- Command — objectives, safety, staff
- Operations — T-card board, assignments
- Planning — IAP tracker, resource status
- Logistics — ICS-205 comms plan, supplies
- Finance/Admin — cost, time, procurement
- Planning P — 15-phase planning cycle guide
- ICS-213 / ICS-214 / ICS-309 forms
- Winlink Form Import — XML to incident
- Incident types: Disaster, HazMat, SAR, MCI
- Multi-user — all sections see live data

## Connectivity — Dual WAN with Automatic Failover

### InstyConnect Cellular (Primary WAN)

InstyConnect Drum omnidirectional antenna + Switchblade directional backup. T-Mobile + Verizon dual-carrier. Pauses at $5/month between activations.

### Starlink Satellite (Automatic Failover)

Automatic failover when cellular drops. ASUS RT-BE58 Go manages failover with no operator action required. All FieldComms features remain active during satellite operation.

### AMPRNet / 44Net (Amateur Radio IP)

Dedicated Raspberry Pi 5 gateway. WireGuard tunnel to amprgw.ampr.org. Routes 44.0.0.0/8 for all EMCOMM-NET devices. AMPRNet allocation required from portal.ampr.org.

## Hardware Platform

| COMPONENT | SPECIFICATION | ROLE |
|---|---|---|
| FieldComms Server | Pi 5 16 GB · Pironman MAX 5 · 2× 1 TB NVMe RAID 1 | 32 pages · 15 services · FCC DB · Kiwix · maps · Pat |
| 44Net Gateway | Pi 5 16 GB · Argon NEO 5 · 256 GB SSD · Pi OS Desktop | WireGuard to AMPRNet · routes 44.0.0.0/8 · status at :9000 |
| Wi-Fi + Switch | ASUS RT-BE58 Go Wi-Fi 7 · UniFi Lite 16 PoE (16-port GbE) | EMCOMM-NET · DHCP · dual WAN failover · wired hub |
| Primary WAN (Cellular) | InstyConnect Drum (omni) + Switchblade (directional) · 5G/LTE | T-Mobile + Verizon · pause at $5/month · swap in 5 min |
| Secondary WAN (Satellite) | Starlink dish + Ethernet adapter → ASUS USB WAN | Auto-failover when cellular drops · all features stay active |
| Radio + Workstations | IC-7300 · Winlink · VARA HF · JS8Call · Pi 500 ×4 · Monitor ×4 | HF comms · browser operator stations · no app install |

### Reference & Administration

| | |
|---|---|
| **Kiwix Library** | WikiMed, Wikipedia, iFixit — offline |
| **Reference Library** | SOGs, plans, field docs — searchable |
| **Print Center** | ICS 213/214/309, NTS, access cards |
| **Cheat Sheets** | Phonetics, Q-codes, CTCSS, band plan |
| **Callsign Lookup** | 800K licensees — instant offline |
| **Pre-Flight Check** | GO / CAUTION / NO-GO readiness |
| **WAN Dashboard** | InstyConnect, Starlink, 44Net live |
| **Offline Maps** | USGS topo — APRS and SAR maps |
| **Health Monitor** | CPU, disk, GPS, internet, services |
| **Access Cards** | Avery 5371 printable wallet cards |

### HOW TO ACCESS

**1** Wi-Fi: **EMCOMM-NET**
**2** Browser: **http://192.168.50.1**
**3** Select mode: Amateur / Starcom / ICS

**No app · No login · Any device · Any OS**

| 32 | 15 | 6 |
|---|---|---|
| Web Pages | Services | WAN Sources |

### WHY FIELDCOMMS

| | |
|---|---|
| **Fully Offline** | All 32 tools work with zero internet |
| **Multi-User** | Every operator sees live data simultaneously |
| **30-Min Deploy** | Unpack and operational in under 30 minutes |
| **Dual WAN** | Cellular primary, satellite auto-failover |
| **RAID 1** | Drive failure does not stop operations |
| **Full ICS** | All 5 sections, Planning P, printable IAP |
| **Radio Integrated** | Winlink, JS8Call, APRS, PACTOR, Pat |
| **Grant Eligible** | FEMA BRIC, ARPA-E, ARRL Foundation, DHS |

501(c)(3) Grant Eligible · Funding Sources: FEMA BRIC · ARPA-E · ARRL Foundation · ARES Special Fund · DHS BRIC Nonprofit Security Grant Program · Illinois Homeland Security